Table 1: Comparison of Equation Search and Generation Strategies in AI4Science

Search Strategy	**Mechanism**	**Discovery Speed**	**Complexity Limit**	**Best For**
Pathwise Matrix (FFX)	Sparse deterministic linear regression over fixed libraries	Instantaneous (Seconds)	Low to Medium	Ultra-fast baselines, large datasets
Time-Delay Search (NARX-SR)	Recurrent, lag-aware tree or orthogonal regressions	Fast to Medium (Minutes)	Medium to High	Time-series with feedback loops and historical memory
Physics Decomposition (AI Feynman)	Structural breakdown via physics priors (symmetries, units) + NN fit	Slow (Hours)	High	Uncovering exact analytic laws of physics from raw data
Evolutionary Tree (PySR)	Genetic mutations and crossover of operator trees	Slow to Medium (Mins/Hrs)	High	Open-ended scientific discovery
Sparse Dynamics (SINDy)	Sequentially thresholded ridge regression	Fast	Medium	Physical systems (ODEs/PDEs)
Neural Generation (DSR)	Token-by-token sequence generation via RL	Medium	Medium	Finding specific structural patterns
Agentic Loop (SR-Scientist)	LLM code writing, execution, and optimization	Slow (Agent loops)	Very High	Noisy data needing domain knowledge